



REVIEW ARTICLE OPEN ACCESS

Technical Considerations in the TCD Evaluation of Cerebral Circulatory Arrest in Death by Neurological Criteria

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ABSTRACT

Background and Purpose: Ancillary testing is essential for diagnosing brain death/death by neurologic criteria (BD/DNC) when clinical determination (neurologic exam and apnea testing) cannot be fully or safely performed. Transcranial Doppler ultrasound (TCD) has been recommended as an ancillary test; however, the technical descriptions have been heterogeneous. We aimed to consolidate the technical considerations related to the use of TCD as an ancillary test in the evaluation of BD/DNC.

Methods: We performed a scoping review of the available literature from inception to December 2024. Articles describing TCD as an ancillary test for BD/DNC were included using appropriate Medical Subject Headings terms. The focus was on technical details and clinimetric properties of TCD (e.g., signal quality, frequency, gate, probe location, etc.). Technical categories were curated as full-text reviews were completed.

Results: We reviewed 181 articles to extract technical descriptions of TCD under several categories. Image acquisition was divided into machine-related factors (59 studies, four questions), exam-related aspects (82 studies, 13 questions), and patient anatomical factors (19 studies, three questions). Image interpretation concerns were addressed in 50 studies (three questions). Patient-related clinical factors affecting TCD utility were addressed in 44 studies (seven questions). Legislative priorities were discussed in 19 studies; in 13 studies, authors commented on their recommendations for research priorities.

Conclusion: We found significant inconsistencies and a lack of technical details described in the literature. Next steps include a national survey of current TCD protocols for BD/DNC evaluation, and an expert consensus guidelines report on the technical use of TCD as an ancillary test.

This work was previously presented as an abstract poster at the 22nd Annual Meeting of the Neurocritical Care Society.

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1 | Introduction

Ancillary testing is required to complete the determination of brain death/death by neurologic criteria (BD/DNC) in situations in which the clinical assessment cannot be fully completed or is confounded. This includes hemodynamic or pulmonary instability precluding apnea testing, uncorrectable confounding, inability to complete aspects of the neurologic exam, or uncertainty about possible spinal reflexes. Accepted ancillary tests have varied by country and over time; most recently in the United States, updated 2023 American Academy of Neurology, American Academy of Pediatrics, Child Neurology Society, and Society of Critical Care Medicine guidelines include four-vessel catheter angiography, single-photon emission computed tomography (SPECT) radionuclide perfusion scintigraphy, and transcranial Doppler ultrasonography (TCD) in adults [1]. These three methods are also accepted in the World Brain Death Project, an international effort to provide guidance on BD/DNC [2].

TCD evaluates blood flow in the Circle of Willis and proximal cerebral vasculature. This noninvasive, real-time study can be conducted at the bedside, making it particularly useful for critically ill patients too unstable for travel. Despite recommendations for the use of TCD in guidelines dating back to 1995, its use has been restricted to adults, with considerable heterogeneity in both technical and clinical applications, hindering widespread adoption [3].

This scoping review aims to evaluate published literature with regard to technical considerations for performing and interpreting TCD for the assessment of cerebral circulatory arrest (CCA). Our goal was to identify areas of consensus and discordance to inform guidelines that would standardize clinical application and generalizability.

2 | Methods

This project was the first of a series of projects undertaken by the Brain Death Task force convened by the American Society of Neuroimaging.

2.1 | Eligibility Criteria

To conduct this review, we identified and included all published studies in any language that described the use of TCD in patients for assessment of BD/DNC. Studies were excluded if they exclusively involved any other ancillary tests, such as electroencephalogram (EEG), CT angiography, or radionuclide perfusion scintigraphy. We included both pediatric and adult studies in which TCD was used, despite the lack of guideline endorsement for pediatric use of TCD as an ancillary test since many studies used this to screen patients for CCA. To broaden the search (given the paucity of literature), we did not exclude articles based on study type, language, or population.

2.2 | Review Focus

To consolidate technical details and clinimetric properties of TCD (see Table 1).

2.3 | Search Strategy

The search was designed with the help of a medical librarian to query PubMed and Embase for articles from inception to December 11, 2024. No publication type or language restrictions were applied. To maximize the chance of identifying all relevant studies, in addition to searching databases, we also manually checked the reference lists of the included studies and performed both backward and forward citation analysis.

2.4 | Screening

Screening by title and abstract was conducted by two authors (A.R. and J.S.) independently as described in the Preferred Reporting Items for Systematic reviews and Meta-Analyses flow diagram (Figure 1). The full texts of the included studies were reviewed by three authors (A.R., J.S., and A.H.) for eligibility criteria, and they also performed the citation search and published trial registries. Discrepancies were resolved by consensus.

2.5 | Data Extraction

We extracted any information relevant to technical aspects of performing or interpreting TCD from each full-text article into a master document, which was then reviewed by three investigators (A.S., A.R., and P.U.) and classified into several categories reflecting technical considerations related to image acquisition, image interpretation, patient-related factors affecting TCD utility, legislative priorities, and research priorities. An initial list of technical considerations was generated by the authors, and additional considerations were added as they were identified during full-text reviews. Based on the a posteriori assignment of categories, the technical descriptive content was extracted from selected articles cited to its source. Since the objective of the review was to identify literature to conduct a Delphi-based survey for expert consensus, topics were framed as questions (see Table 1). We tabulated content related to technical consideration classified into three groups: Group 1, where literature irrespective of strength has consistent technical guidance for all aspects; Group 2, where conflicting guidance existed for a given detail; and Group 3, where minimal evidence existed or was published to support or refute a given parameter despite a mention (see Table 2). Each article was also individually reviewed and assigned risk of bias and quality of evidence based on the Oxford Centre for Evidence-Based Medicine (OCEBM) level of evidence (see Table 3) [4]. Broadly, this separates meta-analyses of multiple randomized controlled trials (RCTs), single RCTs, nonrandomized follow-up studies, and case series/reports.

For non-English articles, practicing medical providers in critical care were recruited through personal connections to translate the full text. The volunteer read aloud the full text translating into English, and any technical details were scribed by an author (A.H.). The recording was reviewed to ensure no details were missed.

TABLE 1 | Classification of technical considerations according to technical categories.

Image acquisition—machine-related factors
1. Are there minimal frequency requirements for the probe?
2. What are the minimum gate requirements for pulse Doppler?
3. Are there minimum standards for filter settings?
4. What are the optimal scale settings?
Image acquisition—exam-related factors
5. What is the optimal monitoring duration for each vessel?
6. When should contrast-enhanced ultrasound be considered?
7. Is there an advantage/preference of imaging versus nonimaging TCD?
8. Can power mode only be used to declare CCA?
9. Can color Doppler flow insonation be used to screen for positive diastolic flow?
10. How many vessels need to be insonated for an adequate test?
11. Probe locations (do they need to be specified)?
12. In the absence of temporal windows, are orbital windows an acceptable replacement to evaluate the anterior circulation?
13. Is there a role for insonation of the central retinal artery?
14. In the absence of temporal windows, can carotid Doppler be used to evaluate the anterior circulation?
15. Is there a specific amount of time to wait between the clinical examination for BD/DNC and the use of TCD as an ancillary study?
16. If the first study is consistent with CCA, what is the minimum amount of time to wait until the second study?
17. If the first study is not consistent with CCA, what is the minimum amount of time to wait until another study is performed?
Image acquisition—patient anatomical factors
18. What constitutes a reliable signal?
19. What are the minimum standards to demonstrate absence of windows?
20. If TCD may be used for BD/DNC declaration, should a screening test showing presence of windows be mandated?
Image interpretation
21. What are the technical descriptions of reverberating/oscillating flow and small systolic peaks?
22. Is there a minimum training requirement to perform and/or interpret TCD for diagnosing CCA?
23. Who should be performing and/or interpreting the test?
Patient-related clinical factors affecting utility of TCD for CCA
24. Is there any pediatric population where TCD can be considered?

(Continues)

TABLE 1 | (Continued)

25. Can TCD be used in patients with hemicraniectomy, burr holes, or external ventricular drains?
26. Are there special considerations for patients on mechanical circulatory support?
27. Are there considerations in using TCD in patients with arrhythmia?
28. Are there any other conditions in which TCD may not be accurate?
29. What vital signs should be required and documented?
30. Are there restrictions on what medications can be used before or during TCD studies?
31. Legislative priorities
32. Author-identified research priorities

Note: This table shows the classification of content related to the questions under each category of technical consideration when using TCD to diagnose CCA.

Abbreviations: CCA, cerebral circulatory arrest; TCD, transcranial Doppler.

3 | Results

A total of 419 studies were identified from PubMed and Embase databases (Figure 1); 186 duplicates were removed, leaving 233 records for screening. These were sought for retrieval, and 51 were not able to be retrieved. One hundred and eighty-two reports were assessed for eligibility, and eight were excluded due to being repeated, retracted, unrelated, or untranslated. A team of collaborators translated 21 articles from eight different languages. Ultimately, 11 new studies were included in the final review, leading to a total of 181 articles reviewed.

3.1 | Image Acquisition—Machine-Related Issues

We identified four technical requirements across a total of 59 articles related to TCD image acquisition machine-related issues (see Table 1 for the list of questions). These requirements are discussed below.

1. *Are there minimal frequency requirements for the probe?*

Nonimaging TCD devices use a 1- to 2.5-MHz probe [5–9], with most publications and guidelines documenting the use of a 2-MHz transducer [6, 8–50]. Imaging TCD is performed with a 2- to 4-MHz [8, 13, 49, 51–53] or 1- to 5-MHz [53, 54] phased array transducer. Extracranial arteries for such studies were reported to be insonated with a 4-MHz continuous wave probe [23] or a 5- to 10-MHz linear array probe [13, 55–57]. An infant study documented transfontanellar insonation with an 8-MHz probe [58].

2. *What are the minimum gate requirements for pulse Doppler?*

Nearly all studies and guidelines that discuss technical considerations for TCD in assessment for CCA describe using pulse Doppler with a range-gated device [18, 24, 42, 55, 59].

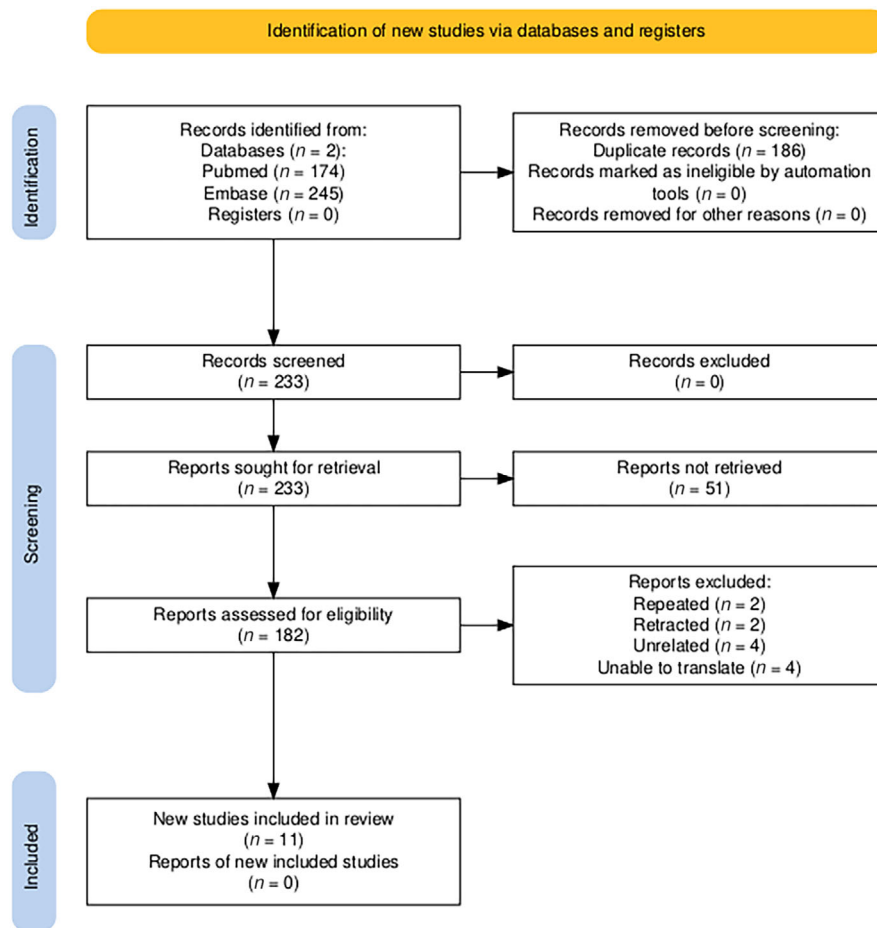


FIGURE 1 | Preferred Reporting Items for Systematic reviews and Meta-Analysis flow chart (*n*, number of studies).

Sample volume varies among studies between 6 mm [46] and 15 mm [6, 33, 42, 60].

3. Are there minimum standards for filter settings?

For the purposes of determining CCA, we found several references recommending high-pass filters to be set to the lowest level [29, 61], commonly cited as ≤ 50 Hz [6, 20, 21, 33, 62, 63], in order to detect low-intensity flow.

4. What are the optimal scale settings?

Some manuscripts recommend lowering the pulse repetition frequency to detect any low-flow signals [6, 52, 57, 64]. Only one [36] guideline highlights using a low scale for color Doppler.

3.2 | Image Acquisition—Exam-Related Issues

We found 13 exam-related issues relating to image acquisition that were discussed in 82 articles; each is described below (see Table 1 for the list of questions).

5. What is the optimal monitoring duration for each vessel?

There was no consensus on this topic. Studies suggest capturing 10 cardiac cycles [65], at least 30 s [33, 40, 59, 66], or at least 3 min [22, 27, 31] with no physiologic data to support these parameters.

6. When should contrast-enhanced ultrasound be considered?

Several publications discuss the use of contrast agents to evaluate patients with seemingly poor temporal bone windows [6, 33, 39, 51, 67]. Two groups [39, 51] describe injecting 2.5 mL of sulfur hexafluoride suspension, followed by a 10-mL sodium chloride bolus (one group suggests injecting via a central venous catheter with the tip at the superior vena cava). An additional 1.5 mL can be injected if the initial bolus is inadequate. Both groups caution that hemodynamic monitoring is required and that patients with unstable ischemic heart disease or severe pulmonary hypertension should be excluded. We did not find any systematic studies evaluating the role of contrast in increasing the sensitivity of TCD as an ancillary test.

7. Is there an advantage/preference of imaging versus nonimaging TCD?

Advantages of transcranial color-coded duplex (TCCD), or imaging TCD, include that it is easier for nonexperts to learn [9, 66, 68] and more practical since the equipment generally already exists

TABLE 2 | Classification of technical considerations according to findings in the literature.

Consistency in the literature	Category of detail	Page number
Are there minimal frequency requirements for the probe	Image acquisition	3
What are the minimum gate requirements for pulse Doppler?	Image acquisition	3
Are there minimum standards for filter settings?	Image acquisition	3
When should contrast-enhanced ultrasound be considered?	Image acquisition	4
Is there an advantage/preference of imaging versus nonimaging TCD?	Image acquisition	6
Probe locations (do they need to be specified)?	Image acquisition	6
If the first study is consistent with CCA, what is the minimum amount of time to wait until the second study?	Image acquisition	8
What constitutes a reliable signal?	Image acquisition	8
What are the minimum standards to demonstrate absence of windows?	Image acquisition	8
Is there a subset of the pediatric population in which TCD can be considered?	Patient-related clinical factors	9
Can TCD be used in patients with hemicraniectomy, burr holes, or external ventricular drains?	Patient-related clinical factors	10
Are there special considerations for patients on mechanical circulatory support?	Patient-related clinical factors	10
Are there any particular conditions in which TCD may not be systematically accurate?	Patient-related clinical factors	10
Inconsistency in the literature	Category of detail	Page number
What is the optimal monitoring duration for each vessel?	Image acquisition	4
How many vessels need to be insonated for an adequate test?	Image acquisition	6
In the absence of temporal windows, are orbital windows an acceptable replacement to evaluate the anterior circulation?	Image acquisition	6
In the absence of temporal windows, can carotid Doppler be used to evaluate the anterior circulation?	Image acquisition	8
Is there a specific amount of time to wait between the clinical examination for BD/DNC and the use of TCD as an ancillary study?	Image acquisition	8
If the first study is not consistent with CCA, what is the minimum amount of time to wait until another study is performed?	Image acquisition	8
What are the technical descriptions of reverberating/oscillating flow and small systolic peaks?	Image interpretation	9
Is there a minimum training requirement to perform and/or interpret TCD for diagnosing CCA?	Image interpretation	9
What vital signs should be required and documented?	Patient-related clinical factors	11
National laws and guidelines	Legislative priorities	11
Sensitivity, specificity, and gold standard comparison	Author-identified research priorities	11
Minimal/no guidance in the literature	Category	Page number
What are the optimal scale settings?	Image acquisition	4
Can power mode only be used to declare CCA?	Image acquisition	6
Can color Doppler flow insonation be used to screen for positive diastolic flow?	Image acquisition	6
Is there a role for insonation of the central retinal artery?	Image acquisition	7
If TCD may be used for BD/DNC determination, should a screening test showing presence of windows be mandated?	Image acquisition	8

(Continues)

TABLE 2 | (Continued)

Minimal/no guidance in the literature	Category	Page number
Who should be performing and/or interpreting the test?	Image interpretation	9
Are there considerations in using TCD in patients with arrhythmia?	Patient-related clinical factors	10
Are there restrictions on what medications can be used before or during TCD studies?	Patient-related clinical factors	11

Note: This table shows the classification of content related to technical consideration according to findings in the literature. Group 1: “Consistency in the literature”—where literature irrespective of strength has consistency on detail. Group 2: “Inconsistency in the literature”—where conflicting guidance existed for a given detail. Group 3: “Minimal/no guidance in the literature”—where minimal/no guidance existed to support or refute.

Abbreviations: CCA, cerebral circulatory arrest; TCD, transcranial Doppler.

in critical care units [52, 68]. Vessels can be more easily and accurately identified due to the visualization of their directionality and imaging of the brain parenchyma [6, 8, 36, 40, 43, 68]. Advantages of nonimaging TCD include smaller size of the transducer and unit, as well as Doppler sensitivity [69]. There is no published preference for the type of machine to use in the determination of CCA.

8. Can power mode only be used to determine CCA?

Two publications [42, 46] discuss nonimaging power M-mode Doppler in addition to traditional Doppler, and neither describes using only power M-mode Doppler in the absence of waveform insonation to determine CCA. We did not find any studies on imaging TCD using power Doppler to enhance the sensitivity of color Doppler.

9. Can color doppler flow insonation be used to screen for positive diastolic flow?

One publication [31] describes the Beacon signal (a pulsating flashing signal due to antegrade and retrograde flow within the same cycle) as being consistently present in CCA. No studies evaluated the feasibility and accuracy of using color Doppler in isolation compared to pulse wave Doppler.

10. How many vessels need to be insonated for an adequate test?

Despite general consensus to insonate both the anterior and posterior circulation, most publications emphasize the need to evaluate more than one vessel [12, 16, 70]. Some manuscripts suggest that the clinical situation may determine the required vessels to be insonated, that is, in cases of BD/DNC from bilateral internal carotid artery (ICA) occlusion, the posterior circulation must be insonated, and in cases of primary posterior fossa injury, the anterior circulation must be insonated [20, 71], while others suggest that the bilateral middle cerebral arteries (MCAs) and some of the vertebrobasilar circulation must all be insonated and consistent [53, 70, 72]. The presence of signals consistent with CCA in the vertebrobasilar circulation is emphasized in several publications [6, 8, 9, 20, 23, 30, 33, 40, 41, 54, 59, 62, 66, 71, 73–75]. As far as specific arterial segments, proximal or distal, there

is some debate about whether the classic TCD waveforms seen in the proximal MCA are adequate or whether the terminal ICA and ICA siphon must also display similar waveforms [76, 77]. Similarly, presence of consistent antegrade flow in either vertebral artery or the basilar artery has been suggested to be inconsistent with CCA [36]. One study has suggested not to record branching points (particularly the MCA–anterior carotid artery branching) as bidirectional flow patterns may be misinterpreted [33]. More recent studies suggest that insonation of the extracranial ICAs, the common carotid arteries, and the vertebral arteries must also be performed [34, 38, 62, 78]. There was no literature specifying the sensitivity of TCD comparing proximal or distal insonation of a vessel or mandating insonating multiple vessel segments.

11. Probe locations (do they need to be specified)?

The transtemporal window is through the squamosal portion of the temporal bone, between the external canthus of the eye and the auricle, approximately 1 cm above the zygomatic arch [5, 7, 9, 10, 15, 21, 25, 40, 52, 57, 79–82]. The suboccipital or transforaminal window is in the midline below the occiput with the transducer angled cephalad through the foramen magnum [5, 7, 9, 35, 42, 52]. The transorbital window can be used to assess the carotid siphon and, at times, the contralateral MCA, and insonation occurs by placing the probe above a closed eyelid [5, 18, 79]. If the extracranial ICA is insonated, this is done via the submandibular window with the probe oriented cephalad [13, 22, 52, 66]. Publications often cited temporal and suboccipital windows but inconsistently.

12. In the absence of temporal windows, are orbital windows an acceptable replacement to evaluate the anterior circulation?

Orbital windows to insonate the carotid siphon have been described when there is an absent temporal window [9, 20, 26, 36, 83]. Using a transorbital approach when transtemporal windows are absent may potentially increase the percentage of patients in whom TCD can be used to evaluate CCA [26, 57, 84, 85], but some raised concerns that routine insonation of the siphon inconsistently distinguishes extracranial from intracranial flow and may actually prevent timely BD/DNC determination if it is inconsistent with the characteristic flow patterns seen with transtemporal and suboccipital windows [40, 66, 86]. It is unclear

TABLE 3 | Level of evidence based on Oxford classification.

Question	Non-randomized controlled cohort	Case series	Case-control	RCT	Meta-analysis	Level of evidence based on Oxford classification
1	10	4	0	0	0	Levels 3 and 4
2	5	0	0	0	0	Level 3
3	3	0	0	0	0	Level 3
4	1	1	0	0	0	Levels 3 and 4
5	1	2	1	0	0	Levels 3 and 4
6	3	0	0	0	0	Level 4
7	0	1	0	0	0	Level 4
8	2	0	0	0	0	Level 3
9	0	1	0	0	0	Level 4
10	13	3	0	0	0	Levels 3 and 4
11	7	2	0	0	0	Levels 3 and 4
12	5	4	0	0	0	Levels 2–4
13	1	2	0	0	0	Levels 3 and 4
14	6	2	1	0	0	Levels 3 and 4
15	5	2	2	1	1	Levels 1–4
16	5	1	0	0	0	Levels 3 and 4
17	2	2	1	0	0	Levels 3 and 4
18	3	0	0	0	0	Level 3
19	2	2	0	0	0	Levels 3 and 4
20	1	1	0	0	0	Levels 3 and 4
21	15	7	1	1	0	Levels 1–4
22	2	0	0	0	0	Level 3
23	0	0	0	0	0	None
24	0	4	0	0	0	Level 4
25	14	5	0	0	1	Levels 3 and 4
26	1	0	1	0	0	Level 4
27	2	1	0	0	0	Levels 3 and 4
28	1	0	0	0	0	Level 3
29	5	2	0	0	0	Levels 3 and 4
30	0	1	0	0	0	Level 4
Total	115	50	7	2	2	

Note: This table presents tabulated information reflecting the Oxford Centre for Evidence-Based Medicine (OCEBM) level of evidence for individual studies in each question [4].

Abbreviation: RCT, randomized controlled trial.

whether insonation of the ophthalmic arteries is useful or valid in diagnosing CCA, as anatomical variations can result in the ophthalmic artery originating extradurally and therefore having seemingly normal flow even when there is intradural CCA [23, 52, 64, 87, 88].

13. Is there a role for insonation of the central retinal artery?

Two publications in adults [64, 87] and one in children [89] describe the insonation of the central retinal artery. Reversal or

absence of diastolic flow can be seen, but waveforms do not necessarily appear to be systolic spikes when diastolic flow drops to zero. It is unclear how reliable this finding is for the evaluation of CCA.

14. In the absence of temporal windows, can carotid doppler be used as a surrogate to evaluate the anterior circulation?

Abnormal waveforms may be seen in the extracranial ICA in CCA. Six studies describe the extracranial ICA displaying systolic

spikes or reversal of diastolic flow [7, 13, 90–93], though at least two of these published exemplary waveforms do not meet the technical criteria by the World Federation of Neurology (WFN) (see below) and show periods of forward diastolic flow. Some studies point out that anastomotic flow through the external carotid artery or ophthalmic artery may preserve some forward diastolic flow despite transtemporal waveforms being consistent with CCA, depending on the existence of ophthalmic and external carotid collateralization [32, 33, 57, 71]. Others describe that extracranial ICA flow can be maintained in patients who otherwise meet criteria BD/DNC [22]. While transcervical insonation of the extracranial ICA in isolation is not a satisfactory ancillary test in any national or committee guidelines, the sensitivity of detecting CCA increases by insonation of the extracranial ICA when no flow can be detected via transtemporal windows [32]. Guidelines do recommend insonation of the extracranial ICA as a complete part of the assessment, but we found no study that compared the sensitivity with or without extracranial insonation.

15. Is there a specific amount of time to wait between the clinical examination for BD/DNC and the use of TCD as an ancillary study?

According to the literature, there is no defined period of time recommended before doing TCD after a clinical examination is consistent with or suspicious of BD/DNC. Several studies suggest that the longer the time, the more likely the TCD will be consistent with CCA [27, 94–96]. Time periods of anywhere from 45 min to 48 h [27, 30, 31, 35, 46, 55, 58, 62, 75, 94, 96, 97] have been described to confirm CCA, though one study suggests that the rates of seeing patterns consistent with CCA plateau by 36 h [30]. CCA may be delayed in which the primary injury is in the posterior fossa [84]. All publications are consistent that the neurological exam for the evaluation of BD/DNC must be done prior to ancillary testing [19, 23, 71, 96].

16. If the first study is consistent with CCA, what is the minimum amount of time to wait until the second study?

The Consensus Opinion by the WFN recommends two separate TCD studies separated by at least 30 min [71], and this standard has been adopted in multiple other guidelines [6, 33, 36, 38, 40, 98, 99]. This standard has also been used in multiple other studies [23, 44, 46, 62, 63, 100, 101], though not all published studies consistently mandate two TCD exams. Some studies suggest a longer period of time [102] between studies, with one study documenting a reversed diastolic flow that reverted back to continuous forward flow after a delay of 150 min [45]. The sensitivity of serial TCD in diagnosing CCA increases with time, but we found no consistent evidence behind the 30-min guidance.

17. If the first study is not consistent with CCA, what is the minimum amount of time to wait until another study is performed?

There is no consensus on how long to wait to repeat a study after the initial one is inconsistent with CCA. Some studies report an

increased sensitivity by repeating TCD every 2–6 h [9, 103, 104], while others recommend waiting at least 12–24 h [27, 31, 75, 105] before repeating the study.

3.3 | Image Acquisition—Patient Anatomical Issues

In our review of patient anatomical issues related to image acquisition, we found three separate concerns that were discussed from a total of 19 manuscripts (see Table 1 for the list of questions).

18. What constitutes a reliable signal?

Studies refer to a reliable signal as being one that can be detected and repeatedly insonated [20, 25]. Publications consistently emphasized that the absence of intracranial flow is not adequate to determine CCA, as this may be due to a lack of acoustic windows. We found several references that mentioned that the absence of flow can be used if there was a previous successful insonation of those same vessels through the same bone window and with the same hemodynamics, and perhaps even by the same technician [25, 52, 59, 60, 78, 106, 107]. When flow is absent, we did not find published evidence of settings needed to show adequate evidence of no flow when previously windows were present and showed oscillating flow or spikes consistent with CCA.

19. What are the minimum standards to demonstrate absence of windows?

It has been reported that a lack of windows can only be determined if there had previously been present flow via the same window under the same clinical and hemodynamic conditions and by the same sonographer [6, 30, 33, 38, 58, 81, 83]. Some studies suggest that two [21, 69] or three [10] experienced independent sonographers must fail to find a signal prior to deeming a window absent. One consensus statement suggests that if a temporal window appears absent, transmission power should be increased and an amplifier should be utilized to increase gain prior to determining absence of a window [36].

20. If TCD is to be used for BD/DNC determination, should a screening test showing presence of windows be mandated?

Given that TCD examinations can provide valuable information on many neurologic disease processes that may lead to BD/DNC (including traumatic brain injury and ischemic and hemorrhagic stroke), it may be reasonable to perform a TCD study early on in such patients [25]. Additionally, in patients with mechanical circulatory support, it may be helpful to obtain a baseline to show cerebral perfusion and underlying cardiac pulsatility for future TCD interpretation [70]. Some national protocols do not allow for absence of signals to be indicative of CCA, even in the presence of a previously found signal; in those countries, a mandatory screening test is not necessary [71]. Despite emerging practices of temporal window insonation and increasing use of TCD for acute brain injury assessment, we did not find any systematic guidance

on performing a TCD early for the purpose of documenting a window in a patient suspected of having or progressing to CCA.

3.4 | Image Interpretation

We identified three questions regarding the interpretation of generated images (see Table 1 for the list of questions) from 50 publications.

21. *What are the technical descriptions of reverberating/oscillating flow and small systolic peaks?*

Reverberating, biphasic, pendular, or oscillating flow is defined as a short antegrade flow during systole, followed by retrograde flow during all of diastole, without any net antegrade flow [6, 9–12, 17, 23, 27, 32–36, 38, 46, 48–50, 54, 70, 71, 74, 82, 94, 101, 107–110]. (The area above and below the spectral envelopes should be equal in the antegrade and retrograde directions.) While most studies consider reverberating flow consistent with CCA when persistent, the exact duration or velocity of systolic flow has not been agreed upon. One study [32] suggests that antegrade flow occupies 15%–50% of the cardiac cycle with retrograde flow taking up 50%–85% of the cardiac cycle. Studies did not specify if diastolic flow needs to be consistently retrograde, though this is generally accepted. Only one study [66] specifically stated that any antegrade flow during the diastolic phase is not consistent with CCA. Adding to the inconsistency, some publications allowed absence of diastolic flow, but with antegrade flow throughout all of systole, to be consistent with CCA [18, 109, 111], and another group [108] suggested that if flow within one cardiac cycle consists of more than two phases, including, for example, a brief period of antegrade flow during diastole or a “triphasic” morphology, it should still be considered “reverberating.” This study [108] suggested that a waveform described as “absence of flow in the diastolic–systolic period” be considered consistent with CCA. In this type of pattern, there was antegrade flow during systole and diastole, but flow returned to zero between systole and diastole, and then again between diastole and the next systole [6, 9–12, 17, 23, 27, 32–36, 38, 46, 48–50, 54, 70, 71, 74, 82, 94, 101, 107–110].

Small systolic peaks or systolic spikes are low-amplitude brief signals that occur in early systole [11, 12, 27, 33, 46, 49, 63, 70, 71, 80, 99, 102, 107, 108, 111, 112] without any antegrade or retrograde flow for the remainder of the cardiac cycle. The highest velocity limit to be considered a spike has been alternatively described as <50 cm/s [6, 9, 29, 32, 33, 36, 43, 49, 54, 66, 71, 100, 105, 106, 113] or <100 cm/s [40, 113], with the duration being <200 ms [2, 5, 29, 44, 54, 59, 91, 95, 99, 106]. Only one study suggested that the time-averaged mean maximum velocity should be less than 10 cm/s [9], while most other studies focused on the highest peak flow velocity of the systolic spike.

22. *Is there a minimum training requirement to perform and/or interpret TCD for diagnosing CCA?*

Given the operator dependence of TCD, some guidelines have commented on the amount of experience required for those performing and interpreting TCD. A Latin American Consen-

sus statement [40] recommended that physicians perform and interpret a minimum of 50 TCD studies in neurocritically ill patients, with at least 20 performed specifically to diagnose CCA. Brazilian guidelines [36] recommend that physicians be certified by the Brazilian Academy of Neurology or complete 200 h of neurosonology training with 12 months of performing TCD/TCCD exams, with additional evidence of 10 supervised exams confirming CCA and 50 supervised TCD assessments. Supervision [36, 114] must be done by a neurosonologist who has been in practice for at least 5 years and be certified by the neurosonology department of the Academy of Neurology. One Chinese study [63] defined trained participants as mid- to senior-level physicians or neurology ultrasound technicians, and a recent Chinese publication described that physicians and technicians performing TCD for DNC/BD determination must have at least 2 years of experience in Brain Death Determination Quality Control Demonstration Hospitals [115]. There are no other guidelines that specifically comment on credentialing or minimum qualifications of personnel performing the TCD examination specifically in the context of the evaluation of CCA.

23. *Who should be performing and/or interpreting the test?*

No guidelines specifically describe who should perform or interpret testing (i.e., technician, clinical physician, or radiologist). Notably, in several Latin American countries, the organ procurement organization is permitted to perform the confirmatory TCD [75, 116].

3.5 | Clinical Factors Affecting the Utility of TCD

We identified seven questions regarding patient-related clinical factors that could alter whether TCD could be confidently used for the identification of CCA (see Table 1 for a list of questions). A total of 44 manuscripts comment on the questions and are discussed below.

24. *Is there a subset of the pediatric population in which TCD can be considered?*

Due to the presence of an open fontanelle in infants, false-negative studies with persistent diastolic flow in a clinically BD/DNC infant have been described [38, 82, 110, 111, 117–119]. Some consensus statements [33, 36, 71] and national laws [116] have stated that TCD is not an acceptable study in those younger than 2 years. The most recent American guidelines [1, 2] on BD/DNC do not allow for TCD to be used as an ancillary test in children. One study was able to replicate classic waveforms seen in CCA with brief fontanelle compression [110]. Other mimics of retrograde diastolic flow that may result in false-positive studies include systemic-to-pulmonary shunts [9], a patent ductus arteriosus [58, 65, 79, 118], or severe aortic valve insufficiency [9, 65], and while this is usually seen in infants, it should be excluded in younger children as well. We did not find any systematic studies outlining the accuracy of TCD in pediatric

patients compared to other ancillary tests despite the increasing use of TCD as a screening test in older adolescents.

25. Can TCD be used in patients with hemicraniectomy, burr holes, or external ventricular drains?

Several guidelines suggest that TCD should not be used, or should be used with caution, in patients with craniectomy, external ventricular drain placement, or skull fractures because of the potential for misinterpretation of preserved blood flow as being inconsistent with BD/DNC [6, 36, 71, 80]. We found several publications that allowed the use of TCD in evaluating CCA in patients with craniectomy. However, we did not find consensus in published literature on whether waveform patterns of reversed diastolic flow, otherwise accepted as consistent with CCA, should also be used as criteria in patients with craniectomy [6, 9, 12, 21, 30, 32, 33, 38, 49, 52, 60, 61, 66, 78, 80, 97, 106, 110, 120–126]. Some publications suggested that preserved diastolic flow on TCD may be a false-negative finding when compared to other ancillary tests like EEG [110]. This has also been discussed in children less than 12 months old with open fontanelles [6, 7, 21, 33, 38, 43, 61]. CCA may still be demonstrated, even in patients with large craniectomies, but the time to diagnosis may be longer in those patients [36, 40, 75, 127]. We did not find consistent evidence of extent of skull defects (burr holes vs. large decompressive hemicraniectomy vs. suboccipital) or interval between craniectomy and TCD performance when evaluating for CCA. Presence of antegrade diastolic flow on TCD can occur in the presence of collaterals in the setting of a surgical decompression; hence, the accuracy of using diastolic flow velocity in these cases is uncertain. Presence of the classic oscillating pattern may help rule in CCA, but antegrade diastolic flow is of uncertain significance.

26. Are there special considerations for patients on mechanical circulatory support?

Given the augmentation of cerebral diastolic flow with the use of intra-aortic balloon pumps (IABPs), diastolic flow may be present in the patient who otherwise might meet criteria for CCA [17, 100, 101, 128], which may be considered a false-negative study. As a result, some argue that the IABP should be placed on standby, or at least the ratio of native cardiac to IABP-assisted beats should be decreased, in order to determine the true circulatory pattern within the brain during native beats [17, 100]. However, this can be challenging due to the underlying cardiogenic shock that necessitates IABP. In continuous assist devices such as venoarterial extracorporeal membrane oxygenation, a native ejection fraction of less than 10% is likely to result in a nonpulsatile TCD waveform [101, 128, 129]. In such a situation, without an IABP to provide pulsatility, classic CCA oscillating patterns will not be seen [52, 129, 130] and only disappearance of TCD signal when it had previously been seen will be the finding consistent with BD/DNC [128, 130–132]. Whether this is acceptable is a matter of contention with varied opinions. Some have found very low continuous flow in clinically BD/DNC patients [17, 130]. We found no systematic evidence related to technical aspects of TCD

in mechanical circulatory support that will add to the accuracy of its use in mechanical circulatory support.

27. Are there considerations for using TCD in patients with arrhythmias?

In severe bradycardia [6], the duration of the systolic spike may extend to longer than 200 ms due to prolongation of overall systole. One manuscript suggests that the heart rate must be >60 during TCD evaluation [52]. Severe tachycardia [6] may make it difficult to assess diastolic flow. There has been concern about changes in TCD waveforms with heart rate in routine clinical evaluation, but we found no systematic mention of considering heart rate when performing TCD for CCA.

28. Are there any particular clinical conditions in which TCD may not be systematically accurate?

False Negatives

There are data to suggest that patients who meet clinical criteria for BD/DNC after anoxic brain injury may not have severely elevated intracranial pressure that would lead to CCA [12, 27, 43]. As a result, false-negative studies may be more likely in that patient population where patients may have a loss of brain and brainstem reflexes seen for a brief period postarrest but preserved cerebral blood flow. Additionally, patients with arteriovenous fistulae [6] or carotid cavernous fistulae [57] may not display the classic findings of [11] CCA because of low-resistance shunting into the venous system causing positive diastolic flow. We also found one reference that showed intact flow in the early stages of intracranial circulatory arrest [11]. Despite the ease of using TCD to check for cerebral blood flow, published literature was lacking in guidance on consistently screening such patients for CCA prior to evaluation for death by neurological criteria.

False Positives

Patients who do not meet clinical criteria for BD/DNC may show reversal of flow during diastole with severe aortic insufficiency [59, 84, 96] or idiopathic hypertrophic subaortic stenosis [84]. Oscillating waveforms may also be seen in high intracranial pressure, and diastolic flow may be restored after institution of intracranial pressure-lowering therapies [103].

29. What vital signs should be required and documented?

The Spanish Society of Neurosonology [33] specifically comments that the partial pressure of CO_2 should be kept between 35 and 45 mmHg and that the mean arterial pressure (MAP) must be ≥ 70 with a blood pressure (BP) no less than 90/50. German guidelines [6] specifically state that blood pressure should be documented and that an MAP >60 must be present for the duration of the TCD study. Brazilian guidelines [36] specifically require a systolic BP

above 90 mmHg, a SpO₂ > 95%, and a heart rate above 60 and recommend the blood pressure be documented. Latin American Consensus [40] guidelines on neurosonology use in BD/DNC recommend an MAP of at least 60 and BP of at least 90/50. Most studies comment on avoiding hypotension as this could result in inadequate cerebral blood flow, but adequate blood pressure is variably defined as blood pressure >90/50 [30, 40, 63, 75], systolic blood pressure >70 [5, 60, 133], systolic blood pressure >100 [1, 52, 62], systolic blood pressure >110 [31], or MAP ≥60 [7, 40, 54, 100], >65 [52], or ≥75 [1]. One study showed that patients with hypotension, acid–base disorders, hypoxia, and hypercarbia can demonstrate triphasic-appearing waveforms that normalize to continuous forward flow for the duration of the cardiac cycle once corrected, suggesting that all metabolic derangements should be corrected prior to TCD acquisition [14]. One study specifically commented that the core body temperature be >32°C [66], while another suggests >35°C [75]. Given the more prevalent use of ancillary testing in patients with confounders, we did not find consistent evidence cited in the guidelines to specify systemic factors that must be met for TCD to maintain accuracy.

30. Are there restrictions on what medications can be used before or during TCD studies?

One study commented that patients undergoing TCD should not have been recently treated with hyperosmolar therapies [103] as enough time should be given for those therapies to have an effect before TCD is performed. Given TCD may often be used as an ancillary test in patients with persistent sedation as a confounder, we found no systematic study evaluating the role of sedation in the accuracy of CCA diagnosis. Cerebral metabolism depressed by sedatives can affect the uptake of tracer by lipophilic agents, but perfusion itself would not be compromised, as TCD only evaluates proximal vasculature, necessitating adequate systemic blood pressure. The nuclear medicine guidelines [134] emphasize the utility of scintigraphy in coma caused by barbiturates but do caution about their effect on decreasing blood flow at high doses. We found no such literature was found for TCD.

3.6 | Legislative Priorities

We found mention of many national laws and guidelines in this review, some of which did not include TCD as an acceptable ancillary test. We found specific literature specifically mentioning the use of TCD for the evaluation of CCA for the following countries: Brazil [36, 75], China [41, 63, 97], Croatia [62], Cuba [95, 135], France [136], Germany [78], Greece [31], India [137], Italy [7, 32, 60, 124, 132], Spain [33], and the United States [1]. Overall, the literature discussed whether TCD was allowable by law, but few provided details regarding the technique or situations in which TCD can be used. One manuscript [98] discussed the disagreement between national protocols regarding whether absence of TCD signals could be used to confirm CCA. Some protocols (Argentina, Ecuador, Germany, Israel) allowed no signal to be considered in the spectrum of consistent finding, while others (Belarus, Malaysia, Moldova, Netherlands, Switzerland, the United States/El Salvador/Kenya/the Bahamas) mentioned using TCD only in the presence of a signal. Mexico's protocol

emphasized the use of TCD in BD/DNC determination only to guide clinicians on when to perform conventional angiography [98].

3.7 | Author-Identified Research Priorities

In reading through existing literature, we noted many author-identified research priorities. The main research questions were regarding sensitivity and specificity. We found wide intervals of reported sensitivity and specificity of TCD for the evaluation of BD/DNC [36, 41, 43, 66, 78, 96, 122, 125, 133, 138, 139]. This likely reflects the variety of standards to which TCD was compared, which include clinical exam with apnea test, digital subtraction angiography, CT angiogram, and EEG. Notably, no studies agreed on the gold standard to which TCD should be compared, and few agreed on the definitions of BD/DNC by other ancillary methods as well, highlighting a research gap in this field [66, 77, 140]. Notably, we did not find robust research on serial evaluation of patients with severe brain injury outlining the chronology and trajectory of different TCD waveforms. We also identified a lack of research on the definitive evaluation of the spectrum of technical specifications for TCD patterns that signify CCA.

4 | Discussion

We uncovered a significant gap in the current literature with regard to detailed technical information for the use of TCD as an ancillary test in evaluating CCA despite increasing interest and a wide consensus for its use to determine BD/DNC as stated in guideline documents from various societies. The German Medical Association has highlighted technical details and differences between other countries' practices [2], while the Latin American Consensus Group derived recommendations from multiple meetings with 26 expert physicians [40]. Despite this, guideline documents do not address technical aspects in multiple areas; for example, the question that addresses the waiting period between clinical examination and TCD (Questions 15, Table 1) to confirm BD/DNC ranges from 45 min to 48 h in the literature. To the issue of the required documented vital signs during a TCD study (Question 29, Table 1), German and Latin American Consensus Guidelines recommend maintaining an MAP >60, while the Spanish Society of Neurosonology specifies an MAP >70. Similarly, there are multiple discrepancies for the duration of systolic flow, followed by retrograde diastolic flow necessary to determine reverberating flow—despite this definition being generally agreed upon (Question 29, Table 1). This highlights the variations in technical descriptions that commonly describe CCA [28, 57].

These emphasize a research gap and the need for expert consensus-based guidelines to address such details to streamline the use of TCD in the evaluation of BD/DNC. These differences in interpretation and practice can lead to inconsistent determinations of BD/DNC based on TCD findings, create distrust in a valid diagnostic modality, and underscore the urgent need for a methodologically standardized approach to using TCD in BD/DNC.

The review also highlights technical details that are well established and agreed upon. Most articles document or recommend using a 2-MHz transducer (Question 1, Table 1). Numerous studies have provided specific descriptions of probe locations, such as the transtemporal and suboccipital windows (Question 11, Table 1). Several guidelines have also clearly advised against or at least recommended using caution regarding the use of TCD in patients with skull defects, such as craniectomy, external ventricular drain, or skull fractures (Question 25, Table 1). While this review has revealed areas of overlooked technical aspects, it has also confirmed some widely accepted practices.

This review has several noteworthy strengths. First, it exclusively focuses on technical aspects of TCD, an area that has not been comprehensively addressed in the previous literature. The search strategy ensured that all available literature on this topic was included from its inception. This approach made the scoping review broad and elaborate, capturing the full spectrum of technical considerations in the use of TCD for BD/DNC diagnosis. Additionally, the review includes studies published in multiple languages, not just English. This multilingual approach has enhanced our understanding of TCD utilization and also highlighted areas of inconsistency.

When assessing the reviewed literature, a marked study design heterogeneity, with the majority being nonrandomized trial design and case series, and topical variation, based on both quantity and level of evidence, were noted. While the volume of literature examining TCD waveform interpretation, its use in cranial defects, and the number of vessels to insonate is greater than in other topics, only one study among these was a randomized trial. Multiple areas exposed a dearth of literature: TCD credentialing backed by a single observational study, pediatric populations with three case series, and no studies looking at the minimal training necessary to interpret TCD for CCA. Only two topics were backed by randomized study design in total: waveform interpretation and timing of the clinical examination to use TCD for assessment of CCA. Concerning the use of TCD in special populations, only one case series explored restrictions on medications that could affect TCD interpretation, none considered arrhythmia, and only a single case series each addressed the use of imaging and color Doppler-based technologies.

The review has some limitations. A primary challenge was taking into account the evolution of TCD technology over time. Findings from older studies may not apply to current practice, and this disparity can complicate the integration of articles that address technical considerations. An additional limitation was the heterogeneity of included articles, not just in technical aspects of TCD, but also the variety of study designs, populations, and methodologies, making it difficult to reach consistent conclusions. A significant portion of the included studies comprised case reports and case series, often lacking rigorous methodology and statistical power. Furthermore, a large number of publications reflected the opinion of the authors and preferred TCD protocols rather than comparative analyses. These factors should be considered when interpreting the results of this review.

Based on this scoping review, we have initiated a national survey of TCD protocols to identify CCA practices and investigate what is

being implemented at the institutional level. Given the significant lack of technical details identified in the literature, our next step is to undertake a Delphi consensus performed with experts in the field in an attempt to standardize the procedure and the interpretation of images, as well as to prevent the clinical use of studies that may result in false positives or negatives. This consensus will aim to also identify the most important research priorities in the field that should be studied. Last, the validation of technical considerations in the appropriate patient population should be attempted, although we recognize the challenges this may pose.

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Conflicts of Interest

The authors declare no conflicts of interest.

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